

Module B5 — Rise and Fall of the Clockwork Universe

5.1 Newton's Laws of Motion

Newton's first law: An object remains at rest or continues to move at constant velocity unless acted upon by a resultant external force.

Newton's second law: The net force acting on an object is equal to the rate of change of its momentum. For constant mass: $F = ma$.

Newton's third law: When two objects interact, they exert equal and opposite forces on each other. These forces act on different objects, are of the same type, and act along the same line.

$$F = ma \quad (\text{constant mass})$$

$$F = \Delta p / \Delta t \quad (\text{general form})$$

Exam Tip: Newton's third law pairs must be the same type of force (both gravitational, both electromagnetic, etc.) acting on different objects. "The table pushes up on the book and the book pushes down on the table" is a valid Newton's third law pair. "Weight of the book and the normal reaction" is NOT — these are different types of force acting on the same object.

Free Body Diagrams

A free body diagram shows all forces acting on a single object, drawn as arrows from the object in the correct directions with appropriate labels. Only forces acting ON the object are shown, not forces the

object exerts on other objects. These diagrams are essential for resolving forces and applying Newton's second law.

5.2 Kinematics — Equations of Motion

For objects moving with uniform (constant) acceleration in a straight line, the SUVAT equations apply:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

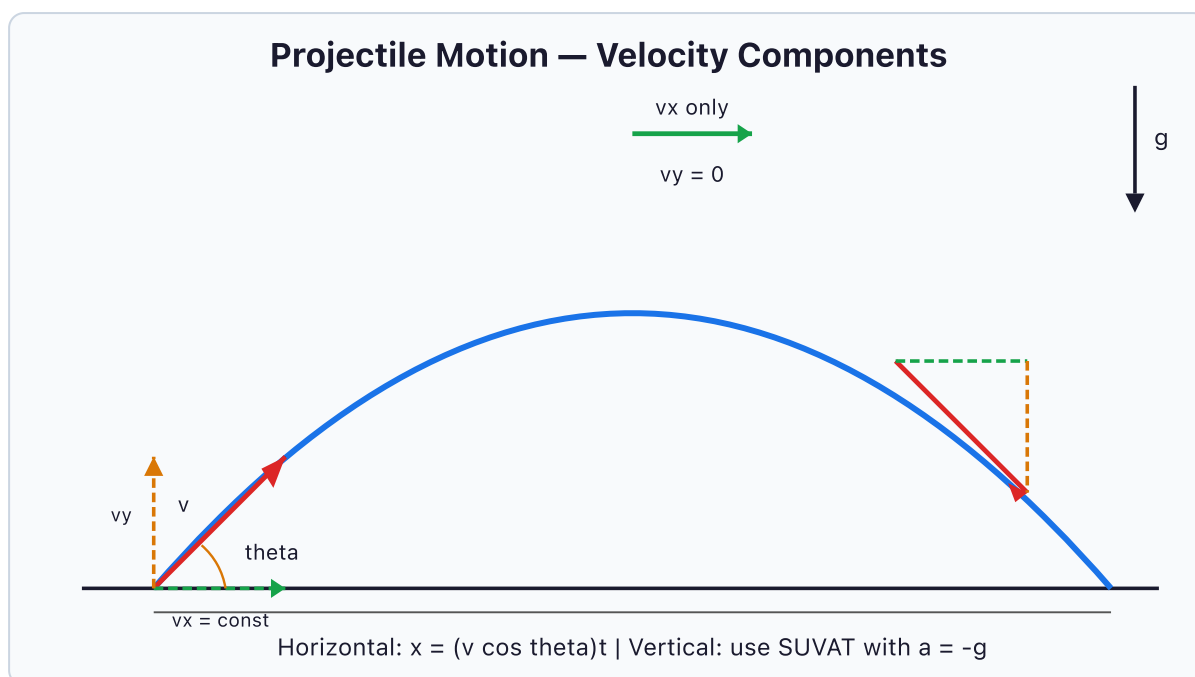
$$s = vt - \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

Symbol	Quantity	Unit
s	Displacement	m
u	Initial velocity	m s^{-1}
v	Final velocity	m s^{-1}
a	Acceleration	m s^{-2}
t	Time	s

Projectile Motion



Projectiles experience only gravitational acceleration (ignoring air resistance). The horizontal and vertical components of motion are independent:

- Horizontally: constant velocity ($a_x = 0$), so $x = u \cos \theta \times t$
- Vertically: uniform acceleration g downward, so use SUVAT with $a = -g$

Exam Tip: For projectile problems, always separate horizontal and vertical motion. List the SUVAT variables for each direction independently. The time of flight is usually the link between the two components.

5.3 Energy

Kinetic energy: The energy an object possesses due to its motion. $E_k = \frac{1}{2}mv^2$

Gravitational potential energy: The energy an object possesses due to its position in a gravitational field. Near the Earth's surface: $E_p = mgh$

Work done: The energy transferred when a force causes a displacement. $W = Fs \cos \theta$, where θ is the angle between the force and the direction of displacement.

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$W = Fs \cos \theta$$

Conservation of Energy

Energy cannot be created or destroyed, only transferred from one form to another. In a closed system, the total energy remains constant. For a falling object (no air resistance), the loss in gravitational potential energy equals the gain in kinetic energy:

$$mgh = \frac{1}{2}mv^2$$

Power and Efficiency

$$\text{Power: } P = W / t = Fv$$

$$\text{Efficiency} = \text{useful output energy} / \text{total input energy} \times 100\%$$

Power: The rate of energy transfer or the rate of doing work. Measured in watts (W), where $1 \text{ W} = 1 \text{ J s}^{-1}$.

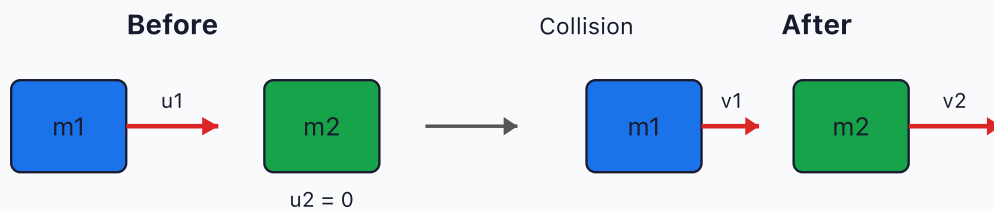
5.4 Momentum

Linear momentum: The product of an object's mass and velocity. $p = mv$. Momentum is a vector quantity, measured in kg m s^{-1} or N s .

Conservation of momentum: In a closed system (no external forces), the total momentum before an interaction equals the total momentum after the interaction.

$$p = mv$$

Conservation of Momentum



$$m1 u1 + m2 u2 = m1 v1 + m2 v2$$

Total momentum before = Total momentum after

Momentum is ALWAYS conserved (no external forces)

Types of Collision

Type	Momentum	Kinetic Energy	Example
Perfectly elastic	Conserved	Conserved	Collisions between gas molecules, hard sphere collisions
Inelastic	Conserved	Not conserved (some converted to other forms)	Most real-world collisions
Perfectly inelastic	Conserved	Maximum kinetic energy lost (objects stick together)	Car crash where vehicles join
Explosion	Conserved (total = 0 if system was initially at rest)	Increases (energy input from chemical/nuclear source)	Firework, bullet fired from gun

Impulse

$$\text{Impulse} = F\Delta t = \Delta p = mv - mu$$

Impulse equals the change in momentum. The area under a force-time graph gives the impulse. This explains why crumple zones, airbags, and crash helmets work: they increase the time over which the momentum changes, reducing the average force on the person.

Exam Tip: Momentum is always conserved in collisions (assuming no external forces). Kinetic energy is only conserved in elastic collisions. To check whether a collision is elastic, calculate the total kinetic energy before and after — if they are equal, it is elastic.

5.5 Circular Motion

Centripetal acceleration: The acceleration directed towards the centre of the circular path. It is caused by the centripetal force and continuously changes the direction of velocity without changing its magnitude.

Angular velocity (ω): The rate of change of angular displacement, measured in rad s^{-1} . $\omega = 2\pi f = 2\pi/T$

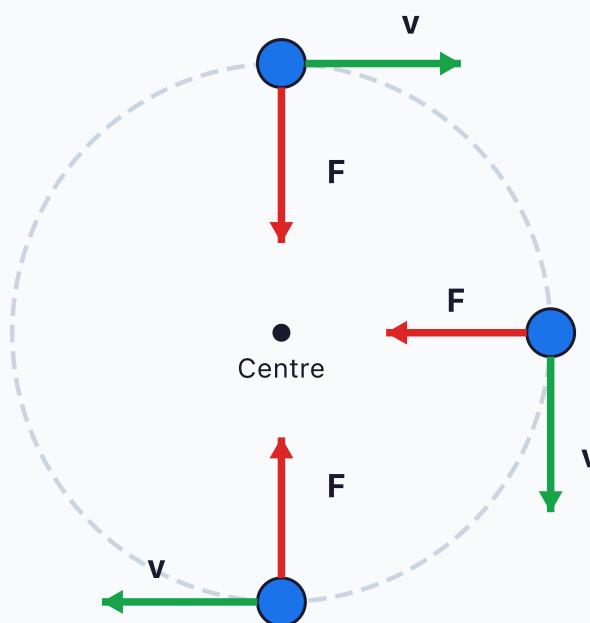
$$v = \omega r$$

$$\text{Centripetal acceleration: } a = v^2/r = \omega^2 r$$

$$\text{Centripetal force: } F = mv^2/r = m\omega^2 r$$

The centripetal force is not a new type of force — it is the resultant of the forces acting towards the centre of the circle. It could be provided by gravity (orbits), tension (conical pendulum), friction (car on a bend), or the normal contact force (clothes in a washing machine drum).

Circular Motion — Force and Velocity Vectors



v is always tangential to the circle
 F (centripetal) always points to the centre

Exam Tip: Never describe centripetal force as a separate force in a free body diagram. It is the net inward force. For example, for a car going over a hill, the centripetal force is the difference between the weight and the normal reaction: $mg - N = mv^2/r$.

5.6 Simple Harmonic Motion (SHM)

Simple harmonic motion: Motion in which the acceleration is directly proportional to the displacement from the equilibrium position and is always directed towards that equilibrium position. $a = -\omega^2 x$

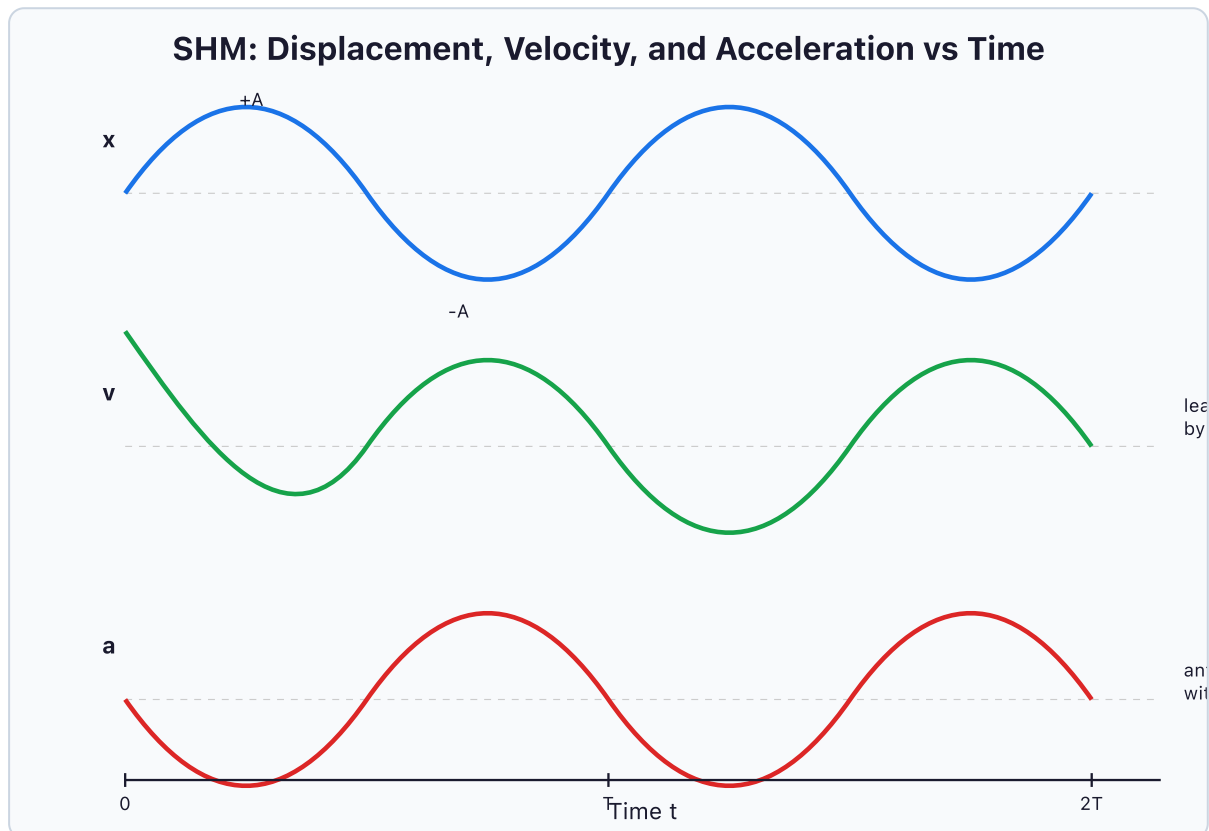
Defining equation: $a = -\omega^2 x$

Displacement: $x = A \cos(\omega t)$ or $x = A \sin(\omega t)$

$$\text{Velocity: } v = \pm\omega\sqrt{A^2 - x^2}$$

$$\text{Maximum velocity: } v_{\max} = \omega A$$

$$\text{Maximum acceleration: } a_{\max} = \omega^2 A$$



SHM Systems

Mass-spring system:

$$T = 2\pi\sqrt{m/k} \quad \text{so} \quad \omega = \sqrt{k/m}$$

Simple pendulum:

$$T = 2\pi\sqrt{l/g} \quad \text{so} \quad \omega = \sqrt{g/l}$$

Note that the period of a simple pendulum is independent of mass and amplitude (for small angles only, typically $< 10^\circ$). The period of a mass-spring system is independent of amplitude and gravity.

Energy in SHM

During SHM, energy continually transfers between kinetic and potential energy. The total energy remains constant (for undamped oscillations):

- At the equilibrium position: E_k is maximum, E_p is minimum (zero)
- At the amplitude: E_k is zero, E_p is maximum
- Total energy: $E_{\text{total}} = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2$

Damping

Damping: The reduction in amplitude of oscillations over time due to energy being dissipated (usually as heat) by resistive forces.

Light damping: The amplitude decreases gradually over many oscillations (exponential decay of amplitude). The frequency remains approximately constant.

Heavy (over) damping: The system returns to equilibrium slowly without oscillating.

Critical damping: The system returns to equilibrium as quickly as possible without oscillating. This is the ideal damping for systems like car suspension and door closing mechanisms.

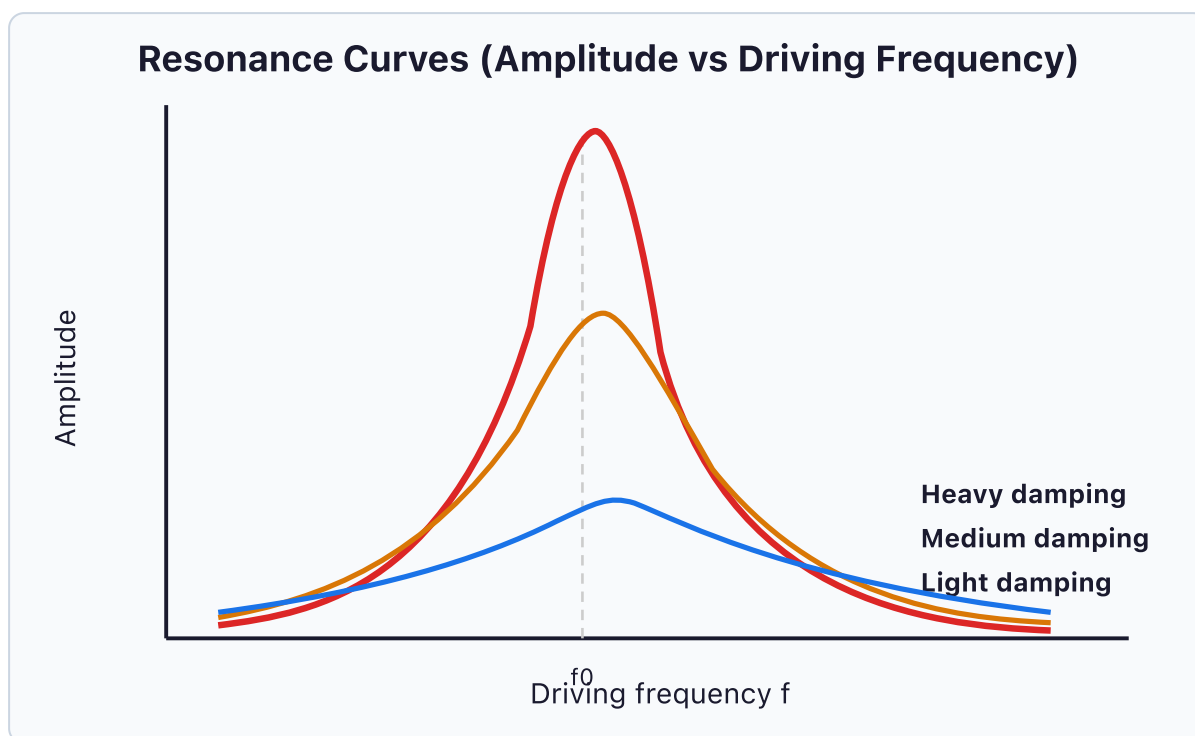
Forced Oscillations and Resonance

Resonance: The dramatic increase in amplitude that occurs when the driving frequency of an external periodic force matches the natural frequency of the oscillating system. At resonance, energy is transferred most efficiently from the driver to the oscillator.

Effects of damping on resonance:

- The peak amplitude at resonance decreases
- The resonance peak becomes broader (less sharp)

- The resonant frequency shifts slightly below the natural frequency



5.7 Thermodynamics

Thermal Energy and Temperature

Internal energy: The sum of the random kinetic energies and potential energies of all the particles in a system. Temperature is a measure of the average kinetic energy of the particles.

Absolute zero: The lowest possible temperature ($0 \text{ K} = -273.15 \text{ }^\circ\text{C}$), at which particles have the minimum possible kinetic energy (zero-point energy in quantum mechanics).

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

Ideal Gas Laws

$$pV = nRT \quad (n = \text{number of moles, } R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1})$$

$$pV = NkT \quad (N = \text{number of molecules, } k = 1.38 \times 10^{-23} \text{ J K}^{-1})$$

The three individual gas laws are special cases:

Law	Constant	Relationship
Boyle's law	Temperature	$p \propto 1/V$ ($pV = \text{constant}$)
Charles's law	Pressure	$V \propto T$ ($V/T = \text{constant}$)
Pressure law	Volume	$p \propto T$ ($p/T = \text{constant}$)

Kinetic Theory of Gases

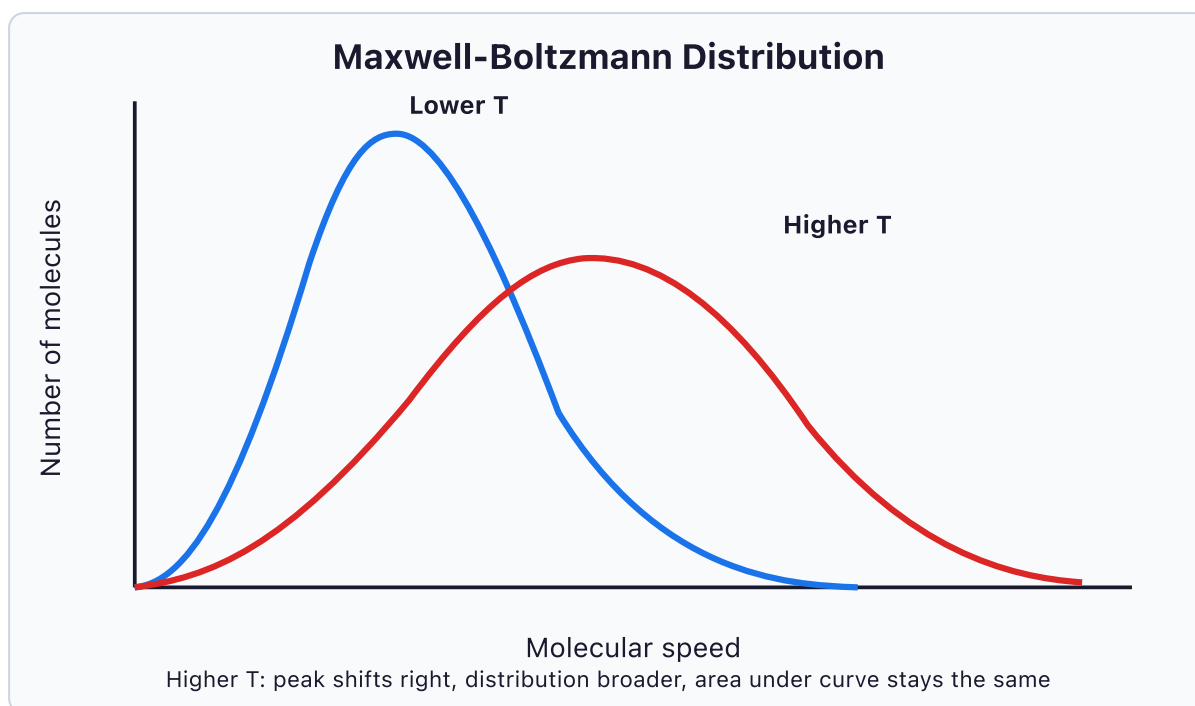
The kinetic theory models a gas as a large number of tiny particles in rapid random motion. Key assumptions include:

- The gas contains a large number of identical molecules
- Molecules move in random directions with a range of speeds
- The volume of the molecules is negligible compared to the volume of the container
- Collisions between molecules and with container walls are perfectly elastic
- The time of a collision is negligible compared to the time between collisions
- There are no intermolecular forces except during collisions

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

$$\text{Average kinetic energy per molecule: } \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

where $\langle c^2 \rangle$ is the mean square speed and k is the Boltzmann constant. The root mean square speed is $c_{\text{rms}} = \sqrt{\langle c^2 \rangle}$.



The Boltzmann Factor

The Boltzmann factor describes the probability that a molecule has energy at least E at temperature T :

$$\text{Probability} \propto e^{-E/(kT)}$$

This explains many temperature-dependent phenomena: the rate of chemical reactions, the proportion of molecules that can escape a liquid (evaporation), thermionic emission, and the electrical conductivity of semiconductors all follow Boltzmann-type relationships.

5.8 Gravitational Fields

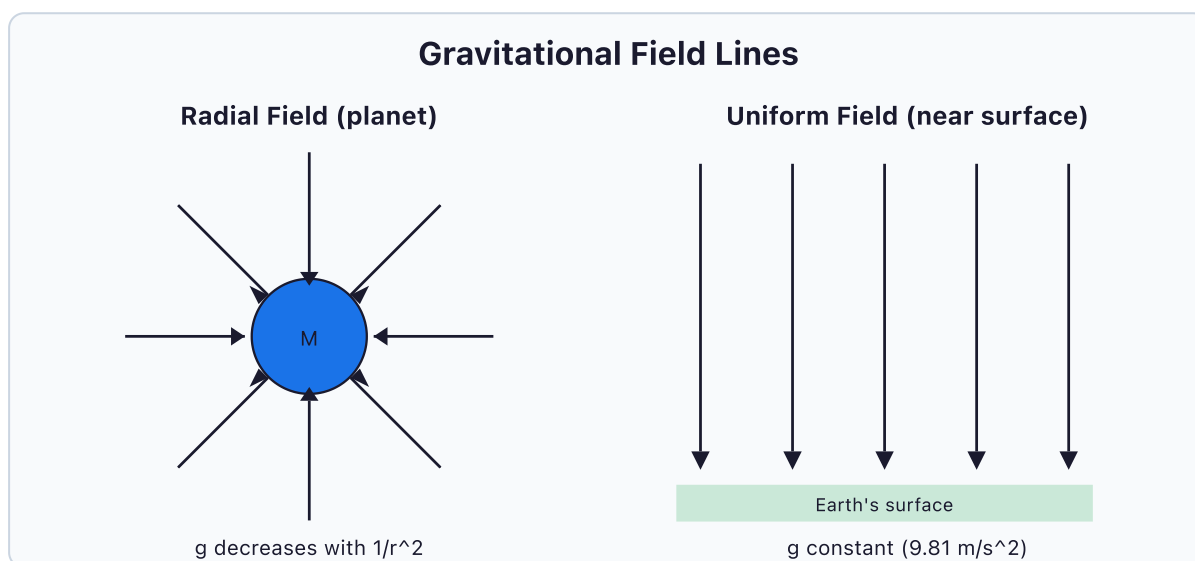
Gravitational field: A region of space where a mass experiences a force due to the presence of another mass. Gravitational field strength at a point is the force per unit mass: $g = F/m$.

Newton's law of gravitation: Every particle attracts every other particle with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = GMm / r^2$$

$$g = GM / r^2$$

where G is the gravitational constant ($6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$), M is the mass creating the field, and r is the distance from the centre of the mass.



Gravitational Potential

Gravitational potential (V_g): The work done per unit mass in bringing a small test mass from infinity to that point. Gravitational potential is always negative (with the zero defined at infinity) because work must be done against the gravitational field to move a mass from the point to infinity.

$$V_g = -GM / r$$

$$\text{Gravitational potential energy: } E_p = -GMm / r$$

Orbital Motion

For a satellite in circular orbit, the gravitational force provides the centripetal force:

$$GMm/r^2 = mv^2/r$$

This gives the orbital speed: $v = \sqrt{GM/r}$. The orbital period can be found from $T = 2\pi r/v$, leading to Kepler's third law:

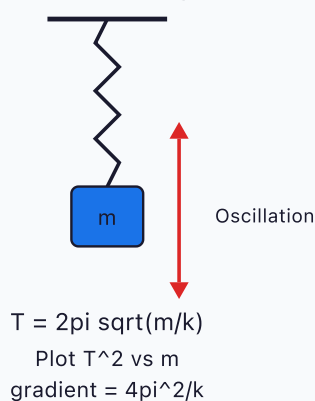
$$T^2 = (4\pi^2/GM) r^3$$

Exam Tip: Remember that gravitational potential is always negative and increases (becomes less negative) as you move away from the mass. The gravitational potential at infinity is zero. Students often confuse gravitational potential (energy per unit mass) with gravitational potential energy (energy of the mass in the field).

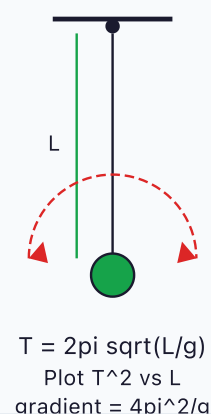
Practical Activity: Investigating SHM (Mass-Spring and Pendulum)

SHM Investigation — Two Systems

Mass-Spring



Simple Pendulum



Apparatus

- **Mass-spring:** Retort stand, clamp, spring, slotted masses, ruler, stopwatch, fiducial marker
- **Pendulum:** Retort stand, clamp, string (~1 m), bob, protractor, ruler, stopwatch

Method

1. For the **mass-spring system**: attach a known mass to the spring. Pull the mass down a small distance and release. Time 10 complete oscillations and divide by 10 to get the period T . Repeat 3 times and take the mean. Change the mass and repeat (at least 6 values).
2. For the **simple pendulum**: measure the length L from the pivot to the centre of the bob. Displace the bob by a small angle (less than 10 degrees). Time 10 complete oscillations and divide by 10 to get T . Repeat 3 times. Change L and repeat (at least 6 values).
3. Mass-spring: plot T^2 (y-axis) against m (x-axis). Gradient = $4\pi^2 / k$, giving k .
4. Pendulum: plot T^2 (y-axis) against L (x-axis). Gradient = $4\pi^2 / g$, giving g .

Safety

- Ensure the retort stand is clamped to the bench or weighted at the base
- Keep amplitude small to ensure SHM approximation is valid
- Stand clear of the oscillating mass

Analysis

Both relationships predict that T^2 is directly proportional to the relevant variable (m or L). A straight line through the origin confirms the SHM relationship. The gradient allows determination of k (spring constant) or g (acceleration due to gravity).

Sources of Error & Improvements

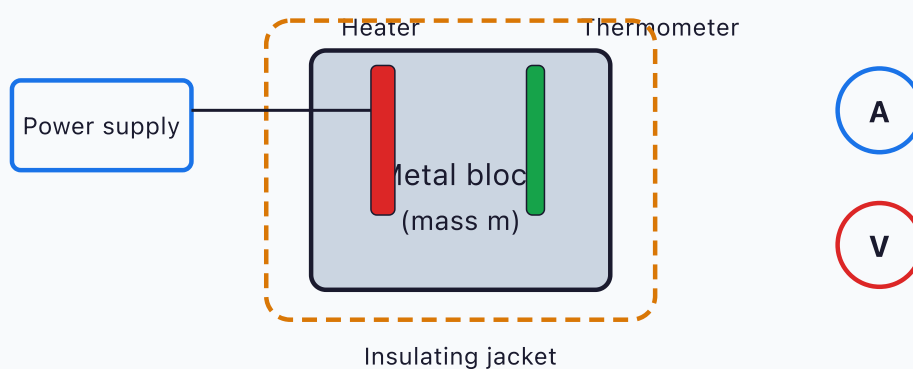
Source of Error	Type	Improvement
Reaction time when starting/stopping stopwatch	Random	Time more oscillations (20+); use a fiducial marker at the equilibrium position
Large amplitude pendulum (not true SHM)	Systematic	Keep angle below 10 degrees for the small angle approximation
Measuring pendulum length to centre of bob	Systematic	Measure to the centre of the bob carefully; use L values that differ by large increments

Exam Tip: Always time at least 10 oscillations to reduce the percentage uncertainty in the period measurement. Counting should start from zero (the first pass of the fiducial marker

is "0", the second is "1", etc.).

Practical Activity: Specific Heat Capacity

Specific Heat Capacity — Electrical Method



$$E = VIt = mc \Delta T, \text{ so } c = VIt / (m \Delta T)$$

Apparatus

- Metal block (e.g., aluminium or copper) with holes for heater and thermometer
- Immersion heater connected to a power supply
- Thermometer (or temperature probe with datalogger)
- Ammeter and voltmeter (to measure electrical energy input)
- Stopwatch
- Digital balance (to measure mass of block)
- Insulating jacket or lagging

Method

1. Measure and record the mass m of the metal block.

2. Insert the heater and thermometer into the block. Add a drop of oil to the thermometer hole for good thermal contact.
3. Wrap the block in insulation to minimise heat loss.
4. Record the initial temperature T_1 .
5. Switch on the heater. Record the ammeter reading I and voltmeter reading V .
6. Heat for a measured time t (e.g., 5-10 minutes).
7. Switch off the heater. Record the maximum temperature T_2 (it may continue rising briefly after the heater is off).
8. Calculate: $c = VIt / (m \times (T_2 - T_1))$.

Safety

- The metal block becomes very hot — handle with tongs or heat-proof gloves
- Do not touch the heater element
- Ensure the heater is fully inserted to avoid it overheating in air

Analysis

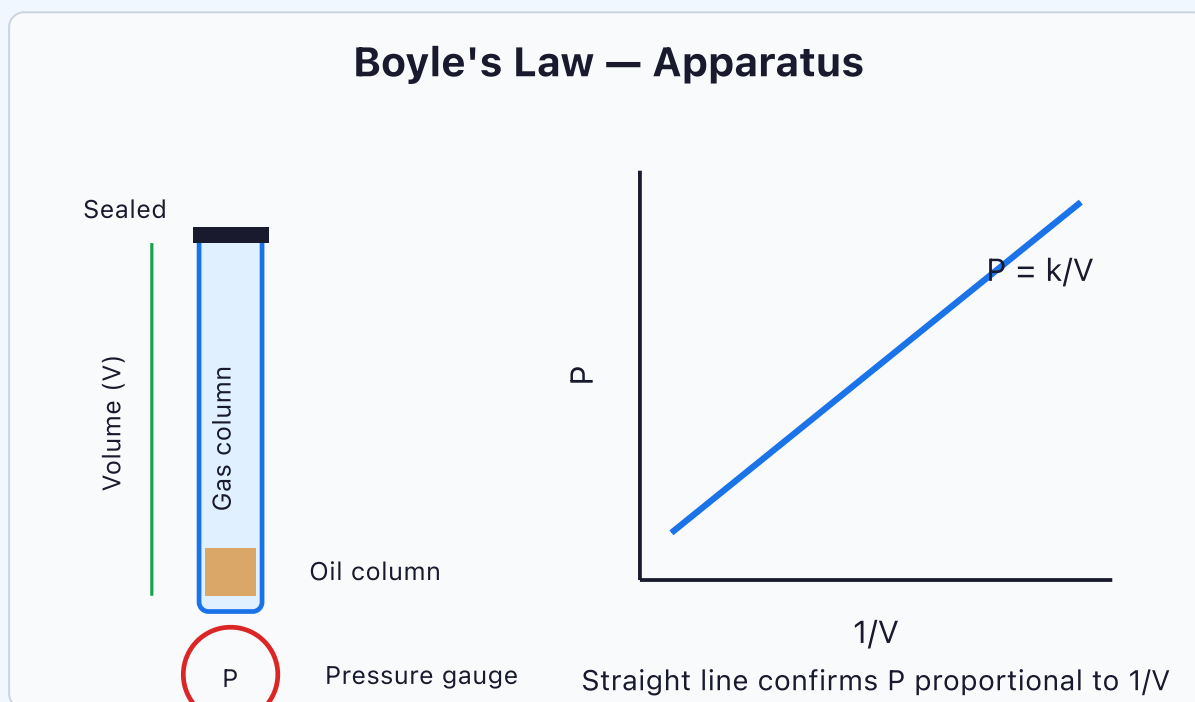
Energy supplied = VIt . Energy absorbed by block = $mc(T_2 - T_1)$. Equating these gives $c = VIt / (m \times \Delta T)$. Compare your result with accepted values (aluminium: $900 \text{ J kg}^{-1} \text{ K}^{-1}$, copper: $385 \text{ J kg}^{-1} \text{ K}^{-1}$).

Sources of Error & Improvements

Source of Error	Type	Improvement
Heat lost to surroundings	Systematic (c appears too large)	Use insulation; start below room temperature and finish above by the same amount
Poor thermal contact between heater/thermometer and block	Systematic	Add a drop of oil to fill air gaps in the holes
Thermometer lag (reading lags behind actual temperature)	Systematic	Wait for thermal equilibrium; read maximum temperature after heater is switched off

Exam Tip: The result for c is almost always an overestimate because some energy is lost to the surroundings. A cooling correction or starting the experiment a few degrees below room temperature can improve accuracy.

Practical Activity: Boyle's Law



Apparatus

- Sealed syringe or Boyle's law apparatus (gas column tube with oil)
- Bourdon gauge or digital pressure sensor
- Ruler (to measure gas column length, proportional to volume)
- Pump or piston to change pressure

Method

1. Record the initial pressure P and volume V of the trapped gas (volume can be read from the syringe scale or calculated from column length and cross-sectional area).

2. Change the pressure by pushing the syringe plunger or adding/removing weights on the piston.
3. Wait for the temperature to equilibrate (the gas warms when compressed) before recording the new P and V .
4. Repeat for at least 6 different pressures.
5. Plot P (y-axis) against $1/V$ (x-axis). A straight line through the origin confirms Boyle's law.
6. Alternatively, plot PV against P — the graph should be a horizontal line.

Safety

- Do not over-pressurise the apparatus — follow manufacturer's guidelines
- If using a glass tube, handle carefully to avoid breakage

Analysis

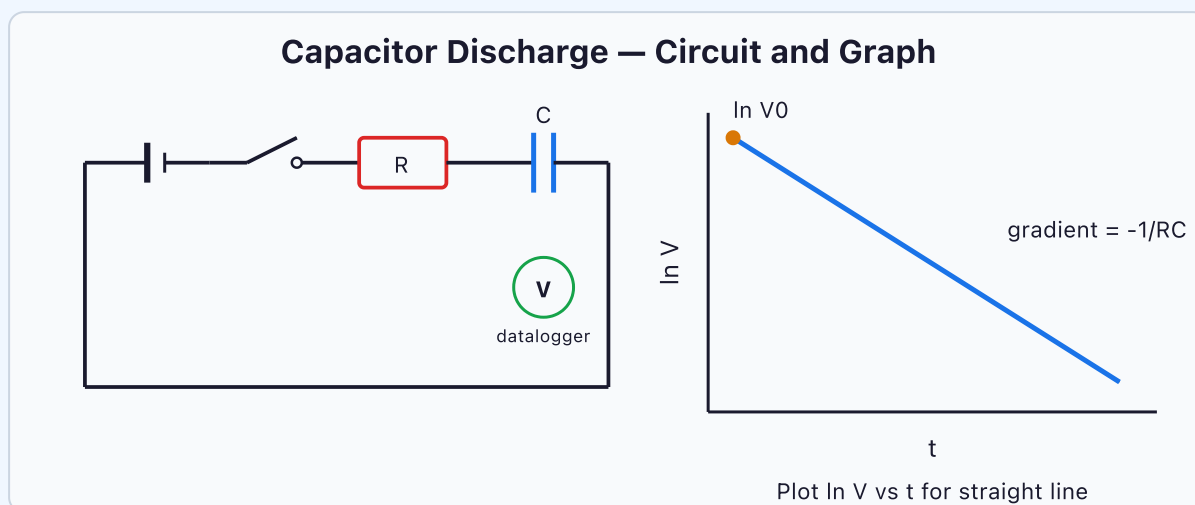
Boyle's law states $pV = \text{constant}$ at constant temperature. Plotting P vs $1/V$ gives a straight line through the origin with gradient = pV (the constant). Plotting PV vs P gives a horizontal line.

Sources of Error & Improvements

Source of Error	Type	Improvement
Temperature rises when gas is compressed	Systematic	Wait after each compression for temperature to return to room temperature
Friction on the piston/syringe	Systematic	Lubricate the piston; tap the syringe gently to overcome static friction
Air leaks from syringe	Systematic	Use sealed apparatus; check for leaks before starting

Exam Tip: The key assumption is constant temperature. If the gas heats up during compression, Boyle's law will not hold exactly. Always allow time for the gas to return to room temperature between readings.

Practical Activity: Capacitor Charge and Discharge



Apparatus

- Capacitor (e.g., 470 μF or 1000 μF electrolytic)
- Resistor (chosen to give a time constant of a few seconds, e.g., 10 k-ohm with 470 μF gives $\tau \sim 4.7 \text{ s}$)
- DC power supply or battery
- Voltmeter or voltage sensor connected to a datalogger
- Switch (two-way if possible, for charge/discharge)
- Stopwatch (if not using datalogger)

Method

1. Charge the capacitor fully by connecting it to the power supply through the resistor.
2. Once fully charged (V reaches V_0), switch to discharge through the resistor.
3. Record voltage across the capacitor at regular time intervals (every 5 or 10 seconds) as it discharges. A datalogger is ideal.
4. Continue recording until V falls to near zero.
5. Plot $\ln(V)$ (y-axis) against t (x-axis). The gradient gives $-1/RC$.
6. The y-intercept gives $\ln(V_0)$, and the time constant $\tau = RC$ can be read from the gradient.

Safety

- Check capacitor polarity — electrolytic capacitors can burst if connected incorrectly
- Use low voltages (below 12 V)
- Discharge the capacitor before modifying the circuit

Analysis

$V = V_0 e^{(-t/RC)}$, so $\ln(V) = \ln(V_0) - t/RC$. The plot of $\ln(V)$ vs t is a straight line with gradient = $-1/RC$ and y-intercept = $\ln(V_0)$. From the gradient, calculate RC and compare with the nominal values of R and C .

Sources of Error & Improvements

Source of Error	Type	Improvement
Voltmeter draws current, affecting discharge rate	Systematic	Use a high-impedance voltmeter or datalogger
Capacitor tolerance (actual C differs from labelled value)	Systematic	Measure RC experimentally from the graph rather than relying on labelled values
Timing errors if using stopwatch manually	Random	Use a datalogger for continuous automatic recording

Exam Tip: The time constant $\tau = RC$ is the time for the voltage to fall to $1/e$ (37%) of its initial value. After $5RC$, the capacitor is effectively fully discharged. You can also find τ by reading the time at which $V = 0.37V_0$ from the exponential decay curve.

Worked Example: A 2.0 kg trolley moving at 3.0 m s^{-1} collides with a stationary 1.0 kg trolley. They stick together. Calculate the velocity after collision and the kinetic energy lost.

By conservation of momentum: $m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$

$$(2.0)(3.0) + (1.0)(0) = (2.0 + 1.0)v$$

$$6.0 = 3.0v, \text{ so } v = \mathbf{2.0 \text{ m s}^{-1}}$$

$$\text{KE before} = \frac{1}{2}(2.0)(3.0)^2 = 9.0 \text{ J}$$

$$\text{KE after} = \frac{1}{2}(3.0)(2.0)^2 = 6.0 \text{ J}$$

$$\text{KE lost} = 9.0 - 6.0 = \mathbf{3.0 \text{ J}}$$
 (converted to heat, sound, and deformation)

This is a perfectly inelastic collision (objects stick together), so KE is not conserved. Momentum is always conserved.

Worked Example: A ball is launched horizontally at 5.0 m s^{-1} from a table 1.2 m high. Calculate the time to hit the ground and the horizontal distance travelled.

$$\text{Vertical: } s = \frac{1}{2}gt^2 \text{ (} u_y = 0, \text{ taking downward as positive)}$$

$$1.2 = \frac{1}{2}(9.81)t^2$$

$$t = \sqrt{(2 \times 1.2 / 9.81)} = \mathbf{0.495 \text{ s}}$$

$$\text{Horizontal: } x = v_x t = 5.0 \times 0.495 = \mathbf{2.5 \text{ m}}$$

The horizontal and vertical motions are independent. The ball takes the same time to fall whether or not it has horizontal velocity.